Training AI to Think Like a Botanist: Using Text-Based Trait Profiles for Interpretable Species Identification in Hyperdiverse Plant Communities

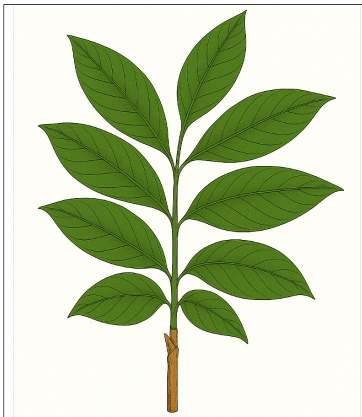
Background

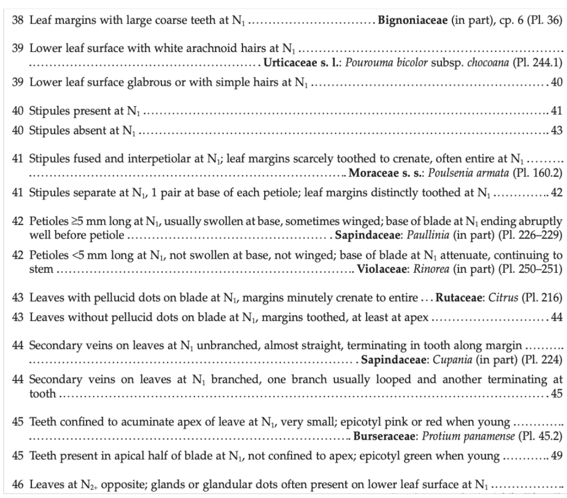
Tropical forests harbor most of Earth's terrestrial biodiversity and provide critical ecosystem services, including climate regulation, carbon sequestration, and water cycling[1,2]. Despite their ecological and economic importance, tropical forests face significant threats from deforestation, habitat fragmentation, and climate change[3,4]. Understanding, monitoring and predicting how tropical ecosystems respond to these challenges fundamentally relies on precise species identification because each species occupies and performs different roles and functions in the ecosystem[5-7]. However, achieving species-level identification remains a major hurdle[8]. Botanists often rely on vegetative morphological traits (e.g., leaf arrangement, margin type, venation patterns, etc.), which, while diagnostic, are often subtle and variable. As a result, field-based identification is time-consuming, difficult to scale, and reliant on highly specialized expertise that is not widely accessible in day-to-day ecological monitoring or restoration work.

This identification challenge is further intensified by the extraordinary species richness found in tropical forests. A few acres of Amazonian rainforest can contain more tree species than all of temperate North America[9]. In such communities, distinguishing among hundreds of co-occurring species, many of which appear remarkably similar, is difficult even for experienced botanists. Current AI approaches often fall short when applied to tropical plant identification[10]. Most rely on convolutional neural networks trained to classify images directly into species categories. While these systems can perform well when trained on large, labeled datasets, they offer limited interpretability and often perform poorly for the many species that are naturally rare and thus likely to be underrepresented in image training sets, a common scenario in hyperdiverse tropical forests. Related efforts using digitized herbarium specimens have leveraged deep learning to annotate traits like phenological states[11], but these models typically depend on preserved, flattened specimens. As such, they do not capture the visual complexity and variability encountered in field conditions and also struggle to generalize beyond their training data. Critically, they tend to ignore the textual morphological traits that form the basis of botanical identification, focusing instead on image-level pattern matching. Therefore, new approaches are needed that can support rapid and accurate species identification, especially for rare taxa and hyperdiverse ecosystems. Such tools are essential for predicting how tropical forests will respond to ongoing environmental changes and for monitoring the success of restoration and conservation efforts.

This proposal builds on insights from a previous seed grant from Yale Planetary Solutions, which explored the use of field photography and hyperspectral data to classify tropical plant species. While that effort has demonstrated the potential of image-based models and yielded promising results for a subset of well-represented species, it also exposed key limitations. Namely, in hyperdiverse forests, many species are naturally rare and thus remain underrepresented in image-based training datasets, constraining model scalability. In contrast, detailed written descriptions exist for a wide range of species and are also widely available across floristic regions. Botanists have long relied on such descriptions, often structured as dichotomous keys, to guide species identification through morphological trait-based logic (Fig 1). This realization inspired our current proposal, which takes a fundamentally innovative approach, one that draws on the rich, underutilized knowledge embedded in botanical texts to teach AI systems to reason like human experts. This new direction also brings together new collaborators with complementary specialized expertise in plant taxonomy, natural language processing, and machine learning infrastructure, expanding the methodological and disciplinary scope of the team beyond that of the previous image-focused project. By focusing on textual morphological trait-based and reasoning-centered models rather than raw image classification, this approach builds on and moves beyond our previous image-focused work. It offers great potential to identify less abundant species to capture the full biodiversity present at a site. Additionally, the proposed approach will be highly scalable because it can be easily adapted to other ecosystems around the world with available floristic texts.

Building on these insights, our project proposes an innovative AI framework inspired by decision-making strategies used by human botanists. Rather than directly classifying images into species, our approach will teach AI systems to recognize diagnostic morphological traits from photographs in the field and use those traits to infer species' identities. We will extract trait information from both existing structured botanical keys and unstructured natural language descriptions found in floras, field guides, and taxonomic monographs. Using natural language processing (NLP), we will convert these prose descriptions into standardized trait profiles that can be paired with tree images for model training. By integrating the logical transparency of traditional botanical identification keys (Fig. 1) with the computational efficiency of modern AI methods, our system aims to provide interpretable, robust, and broadly applicable tools for ecological research. In addition to emulating botanical expert reasoning, this approach may also reveal consistent, visually detectable characteristics that are not yet formally described in existing botanical references, broadening the trait space available for identification. This approach could also help identify species that are potentially new to science by identifying subtle variation in plant traits that may indicate the existence of cryptic species.





38 Leaf margins with large coarse teeth at N_1 .... **Bignoniaceae** (in part), cp. 6 (Pl. 36)

39 Lower leaf surface with white arachnoid hairs at N_1 .... **Urticaceae s. l.**: *Pourouma bicolor* subsp. *chocoana* (Pl. 244.1)

39 Lower leaf surface glabrous or with simple hairs at N_1 40

40 Stipules present at N_1 41

40 Stipules absent at N_1 43

41 Stipules fused and interpetiolar at N_1; leaf margins scarcely toothed to crenate, often entire at N_1 .... **Moraceae s. s.**: *Poulsenia armata* (Pl. 160.2)

41 Stipules separate at N_1, 1 pair at base of each petiole; leaf margins distinctly toothed at N_1 42

42 Petioles ≥5 mm long at N_1, usually swollen at base, sometimes winged; base of blade at N_1 ending abruptly well before petiole .... **Sapindaceae**: *Paullinia* (in part) (Pl. 226–229)

42 Petioles <5 mm long at N_1, not swollen at base, not winged; base of blade at N_1 attenuate, continuing to stem .... **Violaceae**: *Rinorea* (in part) (Pl. 250–251)

43 Leaves with pellucid dots on blade at N_1, margins minutely crenate to entire ... **Rutaceae**: *Citrus* (Pl. 216)

43 Leaves without pellucid dots on blade at N_1, margins toothed, at least at apex 44

44 Secondary veins on leaves at N_1 unbranched, almost straight, terminating in tooth along margin .... **Sapindaceae**: *Cupania* (in part) (Pl. 224)

44 Secondary veins on leaves at N_1 branched, one branch usually looped and another terminating at tooth 45

45 Teeth confined to acuminate apex of leave at N_1, very small; epicotyl pink or red when young .... **Burseraceae**: *Protium panamense* (Pl. 45.2)

45 Teeth present in apical half of blade at N_1, not confined to apex; epicotyl green when young 49

46 Leaves at N_{2+} opposite; glands or glandular dots often present on lower leaf surface at N_1 ....

Fig. 1. Example of morphological trait-based taxonomic key. Diagnostic features such as leaf margins, venation patterns and others are used by botanist to distinguish among major tropical plant groups.

Our ultimate goal is to develop an accessible, interpretable AI platform that empowers researchers, educators, students, and practitioners to accurately identify tropical plant species, significantly enhancing our capacity to monitor, understand, and manage biodiversity in these critical ecosystems. By doing so, we also aim to contribute to advancements in reasoning-centered AI models by grounding model logic in expert botanical knowledge, helping shape transparent, adaptable tools with broader cross-domain relevance. Our objectives therefore are to:

- Identify diagnostic morphological traits (e.g., leaf arrangement, leaf type, margins, venation) that are commonly used in plant species identification and clearly defined in existing botanical resources.
- Curate and expand an annotated image dataset of tropical plant species, including less abundant and rare taxa.
- Pair tree images with structured, text-based species-level trait profiles and determine which morphological traits are most reliably visible in photographs and consistently documented across botanical references (*e.g.,* Smithsonian Tropical Research Institute species database, digital floras, and taxonomic guides).

- Train and evaluate reasoning-centered AI models capable of interpreting morphological traits to infer species identity, emphasizing model accuracy, interpretability, and applicability in field conditions.
- Compare the performance of multiple model architectures, including convolutional neural networks (CNNs) and multimodal models (e.g., BioCLIP), to identify approaches best suited for trait-based detection and transparent scalable species identification.

Methodology

Trait Framework Development and Data Integration: We will develop a framework for morphological traits using comprehensive botanical references. This framework will integrate information from the Smithsonian Tropical Research Institute (STRI) species databases, digital flora, field guides, and other taxonomic resources focused on tropical tree species. Our emphasis will be on traits commonly used by botanists to identify non-reproductive plants, such as leaf arrangement (alternate or opposite), leaf type (simple or compound), margin type (entire, serrated, or lobed), venation patterns, and related features. In addition to structured trait tables, we will extract information from unstructured natural language descriptions using natural language processing (NLP) techniques, enabling us to parse and standardize text-based trait descriptions found in floras and monographs. To ensure consistency across sources, we will harmonize trait terminology and resolve taxonomic inconsistencies through expert review, working closely with botanists familiar tropical floras. The resulting framework will mirror the structure of an advanced botanical key and be readily integrated into machine learning pipelines. Using custom scripts, we will generate species-level trait profiles and link them to corresponding plant images (below) for model training.

Image collection and annotation: We will leverage ongoing data collection, led by Prof. Comita, from a network of long-term forest research sites in central Panama that span gradients of rainfall, soil nutrients, and forest age[12,13]. At each site, tree seedlings and saplings have been censused annually in 400 1-m^2 plots since 2013. Over 12,000 individuals have been tagged, monitored, and identified to species by expert botanists, providing a uniquely robust dataset for understanding tropical plant diversity. We currently have taken ~3,500 photos of these marked plants in the field, organized as an iNaturalist project, with more images to be collected over the course of this project. We aim to expand the dataset to include >10,000 images of >150 species by continuing field photography in Panama and collaborating with other tropical forest researchers. This curated and taxonomically verified image set directly links to written technical species descriptions, offering an ideal foundation for training interpretable AI models that detect key morphological traits and support tropical species inference.

AI Model Development for Trait-Based Inference: We will develop a reasoning-centered workflow that integrates convolution neural networks (CNNs) for image recognition and a fine-tuned contrastive language-image pre-training (CLIP) model for mapping textual descriptions and visual features, to achieve precise morphological trait detection and taxonomic classification of tropical plant images. A CLIP model will be fine-tuned on field condition images paired with curated text-based morphological descriptions and taxonomic labels, establishing robust cross-modal associations within a shared embedding space. In parallel, we will evaluate existing CNNs for high-dimensional feature extraction to capture the visual traits of interest in conventional botanical identification, such as leaf arrangement and margin type and others. CNNs will be evaluated for accuracy using precision, recall, F1-score, and feature visualization review (e.g. Grad-CAM). Extracted feature vectors will be then translated into natural-language descriptions or CLIP-model embeddings; and paired with images to form new text-image inputs. The fine-tuned CLIP model then will be processing new text-image pairs to generate interpretable, biologically grounded predictions.

Advanced Reasoning Models and Chain-of-Thought Integration: To enhance the interpretability and accuracy of our botanical identification system, we will integrate reasoning capabilities of recent large reasoning models, particularly the DeepSeek-R1 family of reasoning models[14.] These models demonstrate advanced capabilities in reasoning through reinforcement learning-based training that naturally emerges with chain-of-thought reasoning patterns, self-verification, and reflection. Our approach will incorporate a multi-stage reasoning pipeline where the model first generates intermediate morphological observations through explicit chain-of-thought processes, then systematically evaluates these observations against botanical knowledge before arriving at species-level conclusions. We will collect expert written chain-of-thoughts that reflect a human thought process in performing classifications: This methodology aims to capture the deliberative process used by expert botanists who methodically examine multiple morphological features, consider alternative hypotheses, and validate their identifications through systematic comparison with reference materials. The reasoning framework will be trained using a reward system that evaluates both accuracy and the biological plausibility of intermediate reasoning steps, ensuring that the model's decision-making process remains grounded in established botanical principles while maintaining the flexibility to adapt to novel morphological variations encountered in tropical forest environments.

Inference-Time Scaling and Computational Optimization: Building on the emerging paradigm of inference-time scaling[15,] our system will implement dynamic computational allocation strategies that allow the model to invest additional processing resources when encountering morphologically complex or ambiguous specimens. This approach will leverage test-time compute power of LLMs, where model performance can be significantly enhanced by generating multiple reasoning paths and employing verification mechanisms to assess and rank outputs by biological accuracy. The system will utilize collaborative verification techniques that combine both chain-of-thought reasoning for interpretability and executable validation pathways for precision, creating a framework for handling the inherent uncertainty in botanical identification tasks. When presented with specimens exhibiting intermediate or unusual morphological characteristics, the model will automatically scale its inference computation, exploring alternative identification pathways, cross-referencing multiple trait combinations, and generating confidence assessments for each potential species match. This inference-time scaling approach is particularly valuable for botanical applications where field conditions, specimen quality, and natural morphological variation can introduce significant uncertainty into the identification process. This modeling strategy will allow us to evaluate the benefits of a trait-centered pipeline in terms of accuracy, interpretability, and generalizability while also aligning closely with the logic of expert taxonomic reasoning.

Results and Broader Impacts: The main expected outputs of this project are 1) A multimodal reasoning-centered AI model capable of inferring species identity by detecting diagnostic morphological traits and simulating expert botanical logic through chain-of-thought reasoning, self-verification, and inference-time scaling. 2) A curated dataset of annotated tree photographs 150+ species linked to structured trait profiles that includes rare species 3) A natural language processing pipeline to extract and standardize traits from floras, field guides, and monographs 4) An interactive prototype tool for field-based identification, supporting targeted follow-up prompts and user feedback in natural language format.

A major advantage of our trait-based approach is its ability to extend beyond common and highly abundant species in the training data. Because predictions are grounded in diagnostic traits, the model could infer identities for rare and underrepresented species and even flag potentially undescribed taxa and offer classification to higher taxonomic levels (e.g., genus or family) when species-level matches are not found. This is especially valuable in tropical forests, where most species are naturally rare yet play important and distinct ecological roles.[5-7] Also, where photo data is scarce, the system could incorporate synthetic or illustrative imagery to support inference. This framework can also support human-AI collaboration by offering both targeted follow-up prompts, or non-experts can contribute natural language descriptions (e.g., "the leaf narrows to a sharp tip.") facilitating real-time refinement of predictions and continuous model improvement.

Overall, this project advances reasoning-centered, interpretable AI by embedding botanical knowledge into model set up and producing interpretable outputs aligned with expert logic. It addresses a widespread challenge in ecology, limited taxonomic capacity in many tropical regions, which has hindered biodiversity monitoring and ecological analysis[8]. By enabling accurate species identification in the field, the system supports researchers, students, and conservation practitioners, and can be readily adapted to other tropical regions with textual floristic resources. Critically, it can also address common limitations in ecological restoration. Many reforestation projects fail due to poor species-environment matching or repeated use of a few widely known species[16]. By enabling accurate identification and selection of locally appropriate taxa, the system can reduce waste and increase the ecological and economic impact of restoration efforts as well as and inform conservation strategies that directly impact livelihoods, food security, and climate resilience in tropical regions. Furthermore, pairing images with descriptive text will enable more accurate AI-generated botanical illustrations of tropical plant species valuable tools for biodiversity education, science communication, and creative applications in media or virtual environments.

While the ecological and conservation aims of this project are central, this project also aims to contribute to advancing reasoning-centered AI systems. By modeling the logic of botanical identification recognizing diagnostic traits and drawing taxonomic inferences, the project engages with emerging approaches in interpretable, reasoning-centered AI. Specifically, it incorporates advanced techniques such as chain-of-thought reasoning, self-verification, and inference-time scaling to simulate expert decision-making processes. To our knowledge, this is the first system designed to emulate botanical key-based decision-making in AI species identification, offering a novel form of interpretable and collaborative AI for biodiversity science. Such models go beyond classification to engage in structured reasoning, providing a transferable framework for applying reasoning-centered AI in other scientific domains.

Ultimately, our proposal directly aligns with the Envisioning AI at Yale Initiative to support innovative, interdisciplinary projects that expand the lines of AI in service of broader societal and scientific goals. By bringing together expertise in ecology, computer science, and botanical taxonomy we aim to deliver a scalable, interpretable tool with strong potential for expansion, and real-world biodiversity impact, while also contributing to the field of reasoning-centered, human-guided, interpretable AI.

References: **(1)** Gibson, L., et al. (2011). *Nature*. **(2)** Dirzo, R. & Raven, P. H. (2003). *Annual Review of Environment and Resources.* **(3)** Pan, Y., et al. (2011). *Science*. **(4)** Hansen, M. C., et al. (2013). *Science.* **(5)** Petchey, O. L. & Gaston, K. J. (2006). *Ecology Letters.* **(6)** Engelbrecht, B. M. J. & Comita, L. S. 2007. Nature **(7)** Poorter, L., et al. 2008. Ecology **(8)** Wheeler, Q. D. & Valdecasas, A. G. (2005). *Nature.* **(9)** Valencia, R., et al. (1994). *Biodiversity and Conservation.* **(10)** Carranza-Rojas, J., et al. (2017). *BMC Evolutionary Biology*. **(11)** Bateux, Q., et al. (2024). *arXiv.* **(12)** Condit, R., et al. (2013). *PNAS.* **(13)** Browne, L., et al. (2021). *Glob. Change Biol*. **(14)** DeepSeek Research. (2024). *arXiv.* **(15)** Snell, C. V., Lee, J., Xu, K., & Kumar, A. (2025). *ICLR.* **(16)** *Brancalion, P. H. S., et al.* (2021). Sci. Total Environ.